

Reshaping Adolescents' Gender Attitudes: Evidence from a School-Based Experiment in India

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Motivation

- **Gender inequality** is acute in many developing countries, and it is deeply rooted in cultural norms.
- India, and Haryana in particular, shows this clearly:
 - Only 0.80 girls enroll in tertiary education for every boy.
 - Child marriage is common.
 - India has one of the world's lowest female labor force participation rates.
 - The child sex ratio is 1.09 boys per girl nationally and 1.20 in Haryana, the most male-skewed state.
- **Key Question: Can a school-based discussion program make adolescents' gender attitudes and behavior more equal, and do the effects persist?**

Introduction

- Past literature has focused on policy constraints. This paper tests whether discussion and persuasion can change [preferences](#).
- The intervention was a school-based program named Legion of Stars.
- Method → [Randomized Control Trial \(RCT\)](#)
- What did the program do?
 - 45-minute classroom discussions every three weeks for 2.5 school years, for students in grades 7-10.
 - Facts and endorsement of gender equality, reflection on their own and society's views (stereotypes, roles at home, girls' education, women's employment, harassment).
 - Some sessions taught communication skills, such as persuading parents to allow later marriage.

Research Question & Results Preview

- Can a school-based intervention that promotes gender equality change gender attitudes of adolescents?
- **Main outcome:** Gender attitudes measured as an index of survey responses on women's roles and rights (e.g., women working outside the home, more women in politics).
- **Findings:**
 - The program shifted attitudes toward gender equality by 0.18 standard deviations after 3.5 months.
 - Approximately 16% of the initial gender-regressive views were converted into support for gender equality.
 - Two years after the program ended, the effect was still 0.16 standard deviations.

Data and Empirical Strategy

- **Sample:** 314 government secondary schools in 4 Haryana districts (150 treatment, 164 control), about 14,000 students surveyed. Randomization was at the school level.
- **Surveys:** Baseline in 2013-14 (grades 6-7, about 12 years old), end line 1 about 3.5 months after the program ended, end line 2 about 2 years later.
- **Empirical Specification:**

$$Y_{ij} = \beta_0 + \beta_1 \text{Treated}_j + \beta_2 Y_{ij}^0 + \beta_3 X_{ij} + \epsilon_{ij} \quad (1)$$

- Y_{ij} : end line outcome for student i in school j
- Treated_j : equals 1 if school j was assigned to treatment, 0 otherwise
- β_1 : Average treatment effect
- Y_{ij}^0 : baseline analog of the outcome
- X_{ij} : grade-gender and district-gender fixed effects

Effect on Gender Attitudes: Short Run vs. Medium Run

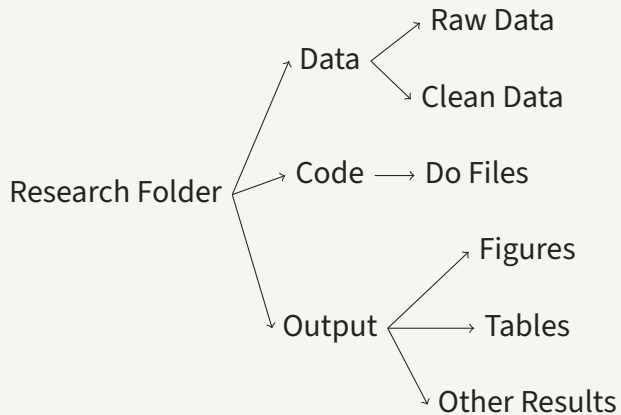
	Gender attitudes index (3.5 months after) (1)	Gender attitudes index (about 2 years after) (2)
Treated	0.180 [0.020]	0.160 [0.019]
Control group mean	0.000	0.333
Basic controls	Yes	Yes
Number of students	13,987	13,679

Notes: Outcome is the gender attitudes index. All regressions control for the baseline analog of the outcome, grade-gender and district-gender fixed effects, and missing flags for each variable used to construct the index. Standard errors in brackets, clustered by school.

Coding/Workflow

- This project uses Stata.
- Everything runs from one script: Master do file → global directories → do files, ado files → work on data.
- Sequence: Baseline 1 cleaning → Endline 1 cleaning → Endline 2 cleaning → Analysis (Tables, Figures).
- Key Packages: outreg2, egenmore, unique, estout/esttab, coefplot, corrtext, putexcel. ([ssc install](#))
- Key Features :
 - Loops: automate repetitive tasks, making the code shorter and reproducible ([foreach](#)).
 - Folder Structure: Data(Rawdata,Cleandata), Code(do files), Output(Figures,Tables,other results).

Folder Structure



Coding/Special Requirements

- **global macros**: the main purpose is to store a folder location once and then reuse it throughout.
- Can be called with **\$name**, e.g.:

```
global main_loc "F:\Exeter_coursework\...\149882-V1"  
global deid "$main_loc/data"  
global baseline_student_raw "$deid/baseline_student_raw"
```

- Write comments to organize do file (*****).
- Requirements:
 - License to Stata or other software.
 - Central High Performance Computing Cluster/Supercomputer.
 - Understanding Stata is strictly **case sensitive**.

README

- What the README does well:
 - Clearly identifies the **master do-file** and explains that it runs the analysis in sequence.
 - Provides a detailed list of the **data, code files, software requirements ,and folder structure.**
 - Clear Diagram of folder workflow.
- What could be clearer:
 - A map in the paper requires **ArcGIS**, but the README does not specify anything on it or provide the underlying step-by-step workflow.

Replication and Conclusion

- The do file was not hard to replicate apart from minor issues such as fixing paths and installing packages.
- The map (districts covering the survey) could not be reproduced.
- The slides are prepared in Latex (Overleaf).
- The results of the paper is important for public policy makers/governments to rethink their approach in schools in India.
- **Takeaway:** Very organized and smooth do file setup.